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Applying local grammars to the diachronic investigation of discourse acts in academic writing: The case of exemplification in Linguistics research articles

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ABSTRACT

This paper proposes a local grammar approach to investigating diachronically discourse acts in academic texts, aiming to offer an additional method for diachronic studies of academic discourse. Local grammars capture the patternings of the realizations of discourse acts at both the lexicogrammatical and discourse-semantic levels; as such they offer a way to explore how the realizations of a given discourse act may vary across time periods or contexts. The proposed local grammar approach is subsequently demonstrated with a study investigating the changing patterns of exemplification in Linguistics research articles, which shows that academic writers of Linguistics nowadays tend to exemplify in simpler and more explicit ways and by presenting subcategories. It is further argued that such investigations complement diachronic investigations of other linguistic features in academic texts, thereby enriching the toolbox of EAP and ESP studies.

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1. Introduction

Diachronic studies of academic discourse have attracted scholarly attention since the 1990s, as "[l]ooking at language change allows us to see the dynamics […] and the contingency of academic writing conventions" (Hyland & Jiang, 2019a, p. 22). Pioneering work in this field includes, for example, Bazerman (1988), Valle (1999), and Atkinson (1999) (see Hyland and Jiang (2019a, pp. 23–27) for an excellent overview of early studies). The most recent and representative diachronic investigations into academic discourse are probably the series of studies by Hyland and Jiang. Using corpora compiled of research articles in Applied Linguistics, Sociology, Electrical Engineering, and Biology, they investigated how engagement (Hyland & Jiang, 2016a), stance (Hyland & Jiang, 2016b, 2018a), (in)formality (Hyland & Jiang, 2017), lexical bundles (Hyland & Jiang, 2018b), citations (Hyland & Jiang, 2018c, 2019b), and metadiscourse (Hyland & Jiang, 2018d, 2020) have changed in academic discourse and how such changes have occurred across these disciplines. Briefly, their studies suggest that the use of stance and engagement expressions have decreased over time (especially in soft disciplines) while citations have significantly increased. They also observe substantial disciplinary variation in such diachronic changes. They find, for example, that writers

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of hard disciplines now use a larger proportion of nominal bundles whereas those of soft disciplines use slightly more research- and text-oriented ones. Despite the significant decrease in the use of interactional metadiscourse in academic discourse in general, hard disciplines have witnessed a substantial increase in interactional features. Overall, their studies have offered important insights into the historical development of academic discourse across disciplines and showcased the value of diachronic studies of academic discourse in general.

Following previous studies, in this paper we present another diachronic study, focusing on the changing patterns of exemplification in academic discourse. The differences between previous studies and the present one lie in two aspects. One is that, while previous studies have investigated diachronically into, for example, metadiscourse, stance, engagement, and lexical bundles, to our knowledge no attempt has yet been made to investigate the diachronic changes of discourse acts (e.g., exemplification, definition) in academic texts. This leads to another more notable difference, that is, the present study is primarily methodological. Specifically, we propose an additional method for diachronic studies of academic discourse, one designed for investigating diachronically discourse acts in academic texts.

The proposed method draws on insights from the concept of local grammar, which refers to an approach to linguistic analysis that seeks to account for specific meanings or functions (Hunston, 2002; Section 3). The term 'discourse act', as used in the present study, corresponds to communicative or rhetorical functions such as definition, exemplification, summarization, and so on. The proposed method is demonstrated with a case study investigating diachronically exemplification in Linguistics research articles. It will be shown that local grammars can be a potentially useful approach to facilitate diachronic investigations into discourse acts in academic writing. It will be argued that such investigations can enrich the description of academic discourse and would also have useful implications for EAP and ESP pedagogy.

The remainder of the paper is organized into four sections. Section 2 discusses exemplification in academic discourse and reviews briefly the few relevant studies published in this area. Section 3 sketches the local grammar approach to diachronic studies of discourse acts in academic writing. It is argued that local grammars capture regularities of language use at both the lexicogrammatical and discourse-semantic levels; as such it offers a way to quantify reliably the realizations of a given discourse act and thereby facilitates diachronic investigation. Section 4 presents the case study, including the corpora used, the local grammar analyses of exemplification, and the results and discussion. Finally, Section 5 concludes the paper, highlighting the usefulness of local grammars in diachronic studies of discourse acts in academic writing and suggesting directions that are worthy of future investigation.

2. Exemplification in academic discourse

Previous research has identified a number of discourse acts that are performed much more frequently in academic texts than in other genres (e.g., Paquot, 2008, 2010). The present study focuses on exemplification, "a communication process through which meaning is clarified or supported by a second unit which illustrates the first by citing an example" (Hyland, 2007, p. 270). The rationale for focusing on this discourse act is twofold. First, exemplification has been found to be a particularly prominent feature of academic writing. Hyland (2007, p. 270), for instance, points out that exemplification is "a recurrent feature of academic writing and a central aspect of exposition". Second, exemplification has been shown to play "a key part in the interactive process between reader and text as the writer anticipates and responds to the reader's possible need for clarification and provides a more accessible way of perceiving it" (Hyland, 2007, p. 278). In other words, with its primary functions to clarify arguments or observations and to provide additional information to illustrate relatively abstract concepts or categories, exemplification is critical to ensuring the persuasiveness and accessibility of academic texts.

While Hyland (2007) focuses on exemplification realized by numerical figures, groups or clauses, he also points out that "example can refer to larger discourse units including case studies" (p. 270). The present study thus considers instances in which existing studies are used for clarification as cases of exemplification as well. Specifically, we define exemplification as a discourse act performed by a writer to illustrate an argument, viewpoint, observation, etc., as in Example (1), or to present subordinate categories or subtypes to illustrate superordinate ones, as in Example (2).

Ex. 1. *It is not surprising that the review articles themselves vary in scope as well as theme. For example, Cazden (1986) provides a very broad-ranging review of sociolinguistic research on classroom discourse.*

Ex. 2. *… ability variables, such as intelligence, aptitude, and cognitive style …*

In Example (1), *For example, Cazden (1986) provides a very broad-ranging review of sociolinguistic research on classroom discourse* illustrates the argument presented in the preceding clause *the review articles themselves vary in scope as well as theme*; and in Example (2), *intelligence, aptitude, and cognitive style* are some specific types of *ability variables*.

Two observations about the use of exemplification in academic discourse are particularly noteworthy. One is that exemplification is typically signaled by a small number of exemplificatory markers. For example, Hyland (2007) examines academic writers' use of several exemplificatory markers such as *an example/instance of*, *for example/instance*, etc. Paquot's (2008, 2010) investigation similarly starts with such markers as *for example*, *for instance*, *illustrate*, and so on. As Hyland and Jiang (2019a, p. 32) note, "to study change means it is necessary to select a set of forms which can be compared

across time". The observation that exemplification is typically marked by a set of lexical markers suggests that it is amenable to corpus-based diachronic investigation.

The other observation is that novice or L2 English writers usually have difficulty using exemplification appropriately. For example, Hinkel (2004, p. 297) observes that "[a]lthough giving examples represents a prevalent explanatory and thesis support strategy in constructing persuasive text in English, teachers and researchers have found that college-level L2 writers rarely employ this strategy successfully and in accordance with the guidelines identified in L2 composition instruction". While this struggle may be partially attributed to variations in the extent to which exemplification is regularly expected among different academic cultures, the primary reason is likely that novice or L2 English writers may have not been well equipped with the lexicogrammatical resources needed to realize exemplification (Paquot, 2008). This points to the need to systematically document the realizations of the discourse act of exemplifying as well as any potential diachronic changes in such realizations in academic discourse. The present study will also attempt to contribute to studies on exemplification in these respects.

A small number of studies have investigated the historical development of exemplifying markers in English, although not in academic contexts. For instance, Rodríguez-Abruñeiras (2012, 2015, 2017) explored the origin and development of various exemplifying markers such as *for example* and *for instance* in English. She found that these two markers were derived from the French nouns *example* and *instance* and that *for example* started to serve the function of exemplifying in the Middle English period and *for instance* in the Early Modern English period. She further suggested that exemplifying markers may have undergone grammaticalization, a process through which certain lexical items or constructions become conventionalized forms for specific grammatical functions in certain contexts (Hopper & Traugott, 2003).

3. Local grammars and diachronic studies of discourse acts in academic texts

This section describes the proposed method for investigating diachronically discourse acts in academic writing, which draws on insights from the corpus-linguistic concept of local grammar (e.g., Cheng & Ching, 2018; Hunston & Sinclair, 2000; Hunston & Su, 2019; Su & Wei, 2018). As an approach to linguistic analysis and explanation, local grammars differ from general or traditional grammars in that each local grammar "seeks to account for, not the whole of a language, but one meaning only" (Hunston, 2002, p. 178). Put more simply, a local grammar "is always a grammar of a discourse function" (Hunston & Su, 2019, p. 571). One defining feature that distinguishes local grammars from general grammars is that local grammars use functionally transparent and context-specific terminologies to analyze instances realizing the target discourse function or act (Bednarek, 2008; Butler, 2004; Hunston & Sinclair, 2000; Su & Wei, 2018). In other words, local grammars attempt to account for the relationship between the syntactic units of an utterance using functional-pragmatic terms rather than traditional grammatical ones such as Subject and Object. For instance, *you* and *readers* in Table 1 indicate the recipient to whom the disclaimers apply and are thus analyzed using the term 'Addressee'. Similarly, *Birmingham* and *cream* in Table 2 refer to the thing being evaluated and are thus labelled 'Target'.

Table 1
Cheng and Ching's (2018) local grammar analyses of disclaiming.

Addressee	Hinge	Degree of warning	Risks
You	are cautioned	strongly	that reliance on any forward–looking statements involves known …
Readers	are cautioned	N/A	that a number of factors could cause actual results to differ …

Table 2
Hunston and Su's (2019) local grammar analyses of evaluation.

Target	Hinge	Evaluation	Specifier
Birmingham	is	famous	as home of the industrial revolution
Cream	is also	helpful	against a dry flaky skin

Before we move further on, a discussion of two methodological issues pertaining to data retrieval and terminology identification is in order. In general, data can be retrieved either by manually identifying instances realizing the target discourse function (Cheng & Ching, 2018) or by a computerized search for a set of grammatical patterns (Hunston & Su, 2019; Su, 2017) or lexical markers (Su & Wei, 2018; Su & Zhang, 2020). In short, data retrieval is largely dependent upon the complexity or conventionalization of the realizations of the target discourse function. The more conventionalized, the more likely a computerized search would work; otherwise, manual identification might be a better option.

Terminology identification may be more challenging, as it requires analysts' cognitive efforts to process the semantic relationship between the syntactic units of the instances realizing the target discourse function. To address this challenge, Su

(2020) suggests two guiding principles. One is termed *The Principle of Transparency*, meaning that the terminologies should be proposed within the specific context of the target discourse function or act; and the other is termed *The Principle of Economy*, meaning that the number of terminologies used to achieve a relatively systematic and comprehensive description of language use should be as small as possible. Su (2020) further argues that, while the two principles may not fully address the challenges associated with terminology identification, they can effectively help guide the identification of terminologies for local grammar research (see Su (2020) for more discussion).

Multiple studies have employed local grammars to account for pragmatic or discourse acts, inter alia, evaluation (Hunston & Sinclair, 2000; Hunston & Su, 2019), definition (Barnbrook, 2002), request (Su, 2017), and disclaiming (Cheng & Ching, 2018). One notable advantage of local grammars, as shown in previous studies, is that they can offer a more transparent and specialized description of the target discourse act than a general grammar can. Barnbrook and Sinclair (2001, p. 273), based on their research on the local grammar of definition, even argue that "such a specialised grammar will outperform a general grammar". In a similar vein, Butler (2004, p. 158) further suggests that "rather than a single general grammar, we might end up with a set of local grammars for particular areas defined by their communicative functions in the discourse". The point, then, is that local grammars constitute a highly useful approach to describing language in use (see also Butler, 2004; Cheng & Ching, 2018; Hunston & Sinclair, 2000; Hunston & Su, 2019; Su & Wei, 2018; Su, 2019).

In addition to their usefulness in language description, local grammars also offer a way to reliably quantify the lexicogrammatical realizations of a given discourse act. This is based on the observation that once the (local) grammar of a given discourse act has been developed, realizations of that discourse act can be identified in a corpus and, accordingly, the frequency of instances realizing a particular local grammar pattern can be calculated (see also Hunston, 2002, pp. 180–181). This has been demonstrated by, for example, Bednarek (2007) who adopts a local grammar approach to compare evaluative language used in broadsheet and tabloid newspapers and finds that the two differ in the frequency of the evaluative patterns used rather than the functions of those patterns. In another study that traces the changing patterns of apology in American English, Su (2020) observes that instances containing the same illocutionary force indicating devices (hereafter IFID) may realize different local grammar patterns and that instances of different IFIDs may realize the same pattern. He thus argues that simply counting the occurrences of given lexical items or IFIDs may not reliably quantify the ways in which a given speech act can be performed and advocates a local grammar approach to diachronic speech act studies.

Following Su (2020), we argue that local grammars can also be applied to explore how discourse acts and their lexicogrammatical realizations have changed over time in academic texts. To test this possibility, we adapted the terminologies used in Su and Zhang (2020; Table 7) and performed a preliminary analysis of instances of exemplification. Like Su (2020), we found that instances containing the same exemplificatory marker may realize different local grammar patterns and that instances containing different exemplificatory markers may realize the same pattern. For example, instances co-occurring with *for example* can realize several patterns, as shown in the set of analyses presented in Table 3.

Table 3
Local grammar analyses of *for example*.

Exemplified	Hinge	Exemplification-RS[a]	Indicator	
… where one or more participants more or less ignore each other's contribution	have been noted by	Sacks	for example.	
Exemplified	Indicator	Exemplification-SC		
… topics impose the need for different communicative skills.	For example,	'Pumps' is an ideal topic for teaching static and dynamic description …		
Exemplified	Indicator	Exemplification-RS		
Outside the USA, there are also some studies on LI retention	(for example,	Hill and Hill 1977 on Aztec in Mexico; Campbell 1980 on Italian in Australia).		
Exemplified	Hinge	Indicator	Exemplification-SC	
… reporting verbs were taken from journals in diverse fields	(including,	for example,	Applied Linguistics, Geology, Public Administration, Engineering)	
Initiator	Indicator	Exemplification-SC		
How do we decide,	for example	that Turn 1 is an O produced by the Teacher as S?		
Exemplified	Hinge	Indicator	Exemplification-RS	
In such cases, it is essential to consider what the target language variety actually is	(see,	for example,	Meisel, Clahsen, and Pienemann 1981).	
Initiator	Exemplified	Hinge	Indicator	Exemplification-RS
For	studies of such errors,	see,	for example	Hatch and Farhady 1982; Borland 1984.

[a] 'Exemplification-RS' stands for 'Exemplification-Relevant study', which means that the act of exemplifying is realized by citing relevant studies; and 'Exemplification-SC' is the short form of 'Exemplification-Subcategory', which means that the act of exemplifying is realized by presenting subcategories (see Section 4.2 for more discussion about this distinction).

Note that in addition to instances co-occurring with *for example*, those co-occurring with other markers can also realize several different patterns, as has been shown in Su and Zhang (2020).

Furthermore, it has also been observed that the same pattern can be realized by instances co-occurring with different exemplificatory markers. For example, our analyses suggest that the pattern Exemplified – Indicator – Exemplification-SC can be realized by instances co-occurring with several markers, as shown in Table 4. In fact, our analyses show that most of the patterns identified can be realized by instances co-occurring with different markers (see Su & Zhang (2020) and Section 4.2 for more examples).

Table 4
The same pattern realized by instances co-occurring with different markers.

Exemplified	Indicator	Exemplification-SC
Some are too general to explain particular errors;	(e.g.	Bernstein's theory of codes, the liberal humanists "expressionism")
We can see the dangers when other researchers resist a coinage;	for instance,	some researchers still do not use the term intron …
… languages vary in terms of the marking of plural concord.	For example,	Hungarian, Turkish, and Basque have nominal affixes for marking number …
Students were required to perform different speech acts	(such as	requesting or giving directions, requesting help in math, or offering to help …)
A more recent expression of this assumption,	illustrated	by a detailed demonstration of how it might be put into practice in ESP …
… whereas the other 12 did not contain a pre-expansion	as exemplified	in Excerpt 1.

It becomes clear that instances co-occurring with the same exemplificatory marker may realize different local grammar patterns and that the same pattern may be realized by instances co-occurring with different markers. The upshot of the above discussion, then, is that simply counting the frequency of exemplificatory markers may not be a reliable way to quantify how exemplification is used and to further explore how it has changed over time. The local grammar approach, on the other hand, could help remedy this problem, as it captures the regularities of both the lexicogrammatical realizations and their discourse semantics (Su, 2020). It is in this respect that we argue that the local grammar approach can facilitate diachronic investigations of discourse acts in academic texts, as will be demonstrated below.

4. The study: changing patterns of exemplification in linguistics research articles

In this section we present the case study. We first discuss the corpora used in the study and how instances of exemplification were retrieved. We then illustrate the local grammar analyses of exemplification and, finally, present the results and discussion. It will be shown that local grammars are a useful approach to diachronic studies of discourse acts in academic writing.

4.1. Corpora and data retrieval

For the subsequent investigation we compiled two corpora by collecting research articles published in the journal *Applied Linguistics* (https://academic.oup.com/applij/). We chose research articles as the academic genre of focus because they are important intellectual and cultural artefacts, relatively easy to access and collect, and have a recognizable stamp of community approval (Hyland & Jiang, 2019a, p. 28). Furthermore, we selected Linguistics as the discipline of focus because writers in this discipline have been shown to use exemplification more frequently than their counterparts in other disciplines (Hyland, 2007). We decided to sample research articles from the journal *Applied Linguistics* because of its relatively long history (i.e., established in 1980), its broad coverage of topics, and its increasing high impact (e.g., it is ranked the second out of 187 journals in Linguistics and has an IF of 4.286, as shown on its official website on April 23, 2021) – properties that can help ensure that the research articles sampled would be largely representative of academic writing in the field. We did not consider the English L1 status of the authors in the sampling process, because all published articles have undergone rigorous peer review and careful editorial processing and as such language use in these articles is acceptable to the disciplinary community.

We focused on two 15-year time periods, one from 1980 to 1994 (TP1), representing Linguistics research articles published in the late 20th century, and the other 2000–2014 (TP2), representing those published in the early 21st century. We purposively left a five-year gap between the two time periods to account for the gradual (as opposed to abrupt) nature of academic discourse evolution (Hyland & Jiang, 2019a). The metainformation of the corpora is given in Table 5.

Table 5
Metainformation of the two corpora.

Time period	No. of texts	No. of tokens	Average length
TP1 (1980–1994)	288	2,717,522	9,435.84
TP2 (2000–2014)	330	4,213,509	12,768.21

Building on previous studies (e.g., Hyland, 2007; Paquot, 2008, 2010; Su & Zhang, 2020), we then identified a set of exemplificatory markers and used them to retrieve instances of exemplification in our corpora. These markers, with their raw and normalized frequency (per million words) in each corpus, are presented in Table 6. A caveat concerning our data retrieval procedure is that, while searching for these typical exemplificatory markers should yield a high hit-rate of instances of exemplification in any corpus, it would nevertheless miss a small number of instances that do not co-occur with any of these predetermined markers. This issue needs to be carefully considered especially for those discourse acts (e.g., describing the methodology, indicating a research gap) that are not consistently signaled by any markers. In such cases, one possible way to remedy this problem is to manually examine a sample of the corpus to ensure the reliability of data retrieval.

Table 6
Exemplificatory markers and their frequency in the two corpora.

Time period	TP1		TP2		LL	LR
Item	Raw	Normed	Raw	Normed		
for example	2,433	895.3	2,363	560.82	260.52**	0.67
for instance,	412	151.61	517	122.7	10.16	0.31
e.g.	1,189	437.53	3,142	745.7	263.33**	−0.77
such as	1,509	555.29	2,716	644.59	21.87**	−0.22
an example of	131	48.21	170	40.35	2.32	0.26
an instance of	17	6.26	36	8.54	1.16	−0.45
exemplify	92	33.85	130	30.85	0.46	0.13
illustrate	459	168.9	885	210.04	14.67*	−0.31
Total	**6,242**	**2,296.95**	**9,959**	**2,363.59**	**3.15**	**−0.04**

Note: LL = log likelihood ratio; ** denotes $p < .0001$; * denotes $p < .001$. The alpha value was adjusted to $p < .0015$ throughout the study as 33 LL tests were run in the study. LR denotes log ratio or the binary log of the ratio of relative frequencies, an effect size measure (Hardie, 2014). An LR score of 1 indicates that an item is twice more common in one corpus than in the other, and each extra point in LR score represents a doubling in size of the difference (e.g., an LR score of 2 indicates an item is four times more common in one corpus than the other).

A glance at Table 6 suggests that the frequency of exemplificatory markers in Linguistics research articles has not changed much overall. Changes in the use of individual markers exist, however. For example, *for example* is used significantly less frequently in TP2 whereas *e.g.*, *such as* and *illustrate* are used significantly more frequently in TP2. Nevertheless, just looking at these numbers probably would not allow us to draw any reliable observations; therefore, below we analyze all the retrieved instances in a more principled manner, with the aim to uncover more systematically the changes between the two time periods in the use of exemplification in Linguistics research articles.

4.2. Data analyses

For the subsequent analyses, we excluded cases that 1) report what other researchers have exemplified, e.g., *Aston (1986) gives an example of how negotiation of meaning by learners of a monolingual background can lead to the negotiated acceptance of* … , 2) use a marker in references, e.g., *Stubbs, M. 1990. Judging the facts: An example of applied discourse analysis* … , 3) use a marker as part of another phrase, e.g., … *look them up in the dictionary for examples*, or 4) use a performative verb in conditional clauses, e.g., … *legal cases assume importance for law students only if they illustrate a particular point of law.* All remaining instances were collaboratively analyzed by the first two authors, and all discrepancies were resolved through discussion.

As discussed in Section 3, local grammars use context-specific functional terminologies to analyze corresponding formal elements. The identification of a set of such terminologies is thus crucial for the subsequent analyses of exemplification. To this end, we adapted the terminologies used in Su and Zhang (2020). The adaptation mainly concerned the distinction made between Exemplification realized by relevant studies and that realized by sub-categories, labelled as Exemplification-RS and Exemplification-SC respectively. Given that academic writers usually exemplify by presenting either relevant studies or sub-categories, as discussed in Section 2, we believe this distinction would help reveal changes related to exemplification in academic discourse. The local grammar terminologies used for the current investigation are presented and explained succinctly in Table 7.

Table 7
Terminologies for analyzing exemplification (adapted from Su & Zhang, 2020).

Terminology	Explanation	Example
Exemplified	The argument, observation or superordinate category/type that is being clarified or illustrated.	Shapira's case is an instance of **the non-learning of English by Zolia.**
Exemplification-RS	Elements that specify or illustrate an argument or observation, usually one or more relevant studies.	… as treated by the ethnomethodologists (for example, Sacks et al., 1974; Turner 1974).
Exemplification-SC	Elements that specify or illustrate a more abstract or superordinate category/type.	… need to unlearn the use of certain language features, such as **the use of tu in formal contexts.**
Indicator	Exemplificatory markers signaling the act of exemplifying.	***For example,*** Oyama (1976, 1978) reported an advantage …
Initiator	Elements that introduce or initiate the act of exemplifying.	***It is shown***, for instance, that this type of phonological …
Researcher	The person who in practice performs the act of exemplifying.	***We*** now illustrate how the above criteria of identifiability …
Hinge	Elements that link different parts.	… these **are** illustrated by examples from the present corpus.
Supporting statement	Elements that provide additional information to clarify or support either the Exemplified or the Exemplification.	For example, Harley (1993) **has shown that French immersion students cannot always distinguish between** …

Overall, the local grammar analyses are largely consistent with those reported in Su and Zhang (2020). The distinction made between Exemplification-RS and Exemplification-SC, however, leads to one major difference in the analyses. Specifically, Su and Zhang (2020) suggest that multi-layered analyses are necessary for instances such as those in Example (3) and (4):

Ex. 3. *Various linguists have criticized the limitations of structuralist semantics. Verschueren, for instance, maintains that structuralism ignores the internal structure of extra-linguistic reality …*

Ex. 4. *Cultural barriers are another problem. For example, Quesada (1976) reports that Mexican-Americans have role expectations which are sometimes at odds with the American …*

Su and Zhang (2020) argue that, at a more abstract level, such instances simply realize the pattern 'Exemplified – Exemplification': the first clause realizes the 'Exemplified' and the second the 'Exemplification'. However, at a more specific level, the second clause realizing 'Exemplification' needs to be further analyzed, as shown in Table 8.

Table 8
Multi-layered analyses in Su and Zhang (2020, p. 7).

Exemplified	Exemplification-RS	Indicator	Supporting statement
Various linguists have criticized the limitations of structuralist semantics.	Verschueren,	for instance,	maintains that structuralism ignores the internal structure of extra-linguistic reality …

Exemplified	Indicator	Exemplification-RS	Supporting statement
Cultural barriers are another problem.	For example,	Quesada (1976)	reports that Mexican-Americans have role expectations which are sometimes at odds with the American …

Su and Zhang (2020) analyses, while acceptable, appear to make the analyses unnecessarily complicated. Using our adapted terminologies, we suggest an alternative analysis, which is presented in Table 9. The revised analyses are both simpler and more comprehensible, showcasing the advantage of the adapted terminologies.

Table 9
Simplified analyses in the present study.

Exemplified	Exemplification-RS	Indicator	Supporting statement
Various linguists have criticized the limitations of structuralist semantics.	Verschueren,	for instance,	maintains that structuralism ignores the internal structure of extra-linguistic reality …

Exemplified	Indicator	Exemplification-RS	Supporting statement
Cultural barriers are another problem.	For example,	Quesada (1976)	reports that Mexican-Americans have role expectations which are sometimes at odds with the …

In addition, we further made a distinction between what we call 'independent exemplification' and 'inserted exemplification'. The former refers to cases where the exemplification is a required component of an utterance and directly affects the semantic completeness of the utterance, as illustrated by Example (5)–(8), whereas the latter refers to cases where the exemplification is embedded within a nominal group and provides illustration or clarification for the exemplified without affecting the semantic completeness of the utterance, as illustrated by Example (9)–(12).

Ex. 5. *An example of this process in action is the emergence of the British Council's language proficiency tests for overseas students.*
Ex. 6. *Shapira's case is an instance of the non-learning of English by Zolia.*
Ex. 7. *The situation is similar when we consider the teaching of related grammatical structures, such as relative clause. For example, Krohn (1971) introduces relative clauses in which the relativized …*
Ex. 8. *These difficulties have to do with potential conflicts between these different criteria. For instance, Paul (1972: 49) reports that natural general is a very strong criterion for gender assignment.*
Ex. 9. *All social control acts, such as directives, fall within this group.*
Ex. 10. *Matching relations, e.g. Compatibility and Contrast, are relations where statements are matched against each other in terms of …*
Ex. 11. *This principle, illustrated in Table 5, is due to the fact that function words …*
Ex. 12. *In the interruption exemplified by Turn 8, the interrupting Speaker addressed the interrupting Speaker.*

The distinction between 'independent exemplification' and 'inserted exemplification' leads to different analyses. Basically, for independent exemplification the entire utterance is considered a semantic unit with the function of exemplifying; consequently, all syntactic units are analyzed using local grammar terminologies; in contrast, for inserted exemplification we only analyze the part of the utterance that realizes exemplification, because the other parts are not necessarily relevant to the function of exemplifying (see the analyses in Table 10).

Table 10
Local grammar analyses of Example (5)–(12).

Indicator	Exemplified	Hinge	Exemplification-SC
An example of	this process in action	is	the emergence of the British Council's language proficiency tests for overseas students.

Exemplification-SC	Hinge	Indicator	Exemplified
Shapira's case	is	an instance of	the non-learning of English by Zolia.

Exemplified	Indicator	Exemplification-RS	Supporting statement
The situation is similar when we consider the teaching of related grammatical structures, such as relative clause.	For example,	Krohn (1971)	introduces relative clauses in which the relativized …
These difficulties have to do with potential conflicts between different criteria.	For instance,	Paul (1972: 49)	reports that natural general is a very strong criterion for gender assignment.

Exemplified	Indicator	Exemplification-SC
All social control acts	such as	directives …
Matching relations,	e.g.	Compatibility and Contrast …
This principle,	illustrated	in Table 5 …
In the interruption	exemplified	by Turn 8 …

4.3. Results and discussion

The analyses above would be sufficient to demonstrate the local grammar analyses of exemplification. Based on the analyses, we identified 37 local grammar patterns of exemplification. To avoid untypical uses, the subsequent discussion focuses on those patterns with 10 or more occurrences in at least one corpus (see the Appendix for the patterns with fewer than 10 occurrences in both corpora). These patterns, along with their examples and quantitative information, are presented in Table 11.

Table 11
Overview of local grammar patterns across TP1 and TP2.

No.	Pattern	TP1		TP2		LL	LR
		Raw	Normed	Raw	Normed		
PA 1	Exemplified – Indicator – Exemplification-SC e.g. *Classifiers are signs for noun categories, such as vehicle, human, animal, flat objects, etc.*	3,581	1,317.74	5,968	1,416.40	11.73*	−0.10

(*continued on next page*)

Table 11 (*continued*)

No.	Pattern	TP1		TP2		LL	LR
		Raw	Normed	Raw	Normed		
PA 2	Exemplified – Indicator – Exemplification-RS e.g. *Other researchers have suggested that such factors should be included in the notion of communicative competence (e.g. Hymes 1972; Jakobovits 1970).*	508	186.94	1,436	340.81	147.32**	−0.87
PA 3	Exemplified – Hinge – Indicator – Exemplification-SC e.g. *The result of this preliminary phase is illustrated in Table 4.*	383	140.94	330	78.32	61.20**	0.85
PA 4	Exemplified – Exemplification-SC – Indicator – Supporting statement e.g …. *when the initiation comes from the writer, the reader is construed as having provided the expected response. A recipe, for instance, can be read without carrying out the instructions.*	294	108.19	284	67.40	32.13**	0.68
PA 5	Exemplified – Hinge – Indicator – Exemplification-RS e.g. *Many studies have concentrated on narrative texts and dialogues (see, for example, House (1984)).*	258	94.94	221	52.45	41.94**	0.86
PA 6	Initiator – Indicator – Exemplification-SC e.g. *What about, for example, re-using someone else's description of background information in writing the ….*	202	74.33	167	39.63	36.28**	0.91
PA 7	Exemplification-SC – Indicator – Exemplified e.g. *Examples 5 and 6 illustrate what happens when play is sanctioned, and perhaps even encouraged, by the teacher.*	159	58.51	374	88.76	20.38**	−0.60
PA 8	Exemplified – Exemplification-RS – Indicator – Supporting statement e.g. *Similar findings were reported by some other SLA researchers. Ranta (2002), for instance, revealed that 'good language learners' in a communicative language teaching environment.*	138	50.78	175	41.53	3.09	0.29
PA 9	Exemplified – Indicator – Exemplification-RS – Supporting statement e.g. *The pattern resembles that of phonological acquisition found in several studies. For example, Jia* et al. *(2006) investigated the effects of time on a large sample of Mandarin …*	96	35.33	205	48.65	6.94	−0.46
PA 10	Exemplification-SC – Hinge – Indicator – Exemplified e.g. *A sample sentence was also provided to illustrate how each item could be used in a sentence.*	90	33.12	129	30.62	0.33	0.11
PA 11	Exemplified – Exemplification-SC – Indicator e.g. *International transcription makes it easier for the user to use other dictionaries (monolingual ones, for example).*	87	32.01	58	13.77	25.48**	1.22
PA 12	Indicator – Exemplified – Hinge – Exemplification-SC e.g. *An instance of how such consistency indices were actually computed is given in Table 1.*	45	16.56	36	8.54	8.81	0.95
PA 13	Initiator – Indicator – Exemplified – Exemplification-SC e.g. *As an example of the procedure in the analysis, a Turkish informant will now be discussed.*	22	8.10	40	9.49	0.37	−0.23
PA 14	Exemplified – Hinge – Exemplification-RS – Indicator e.g. *This and other research (see Bialystok 1979 for example) is strongly suggestive of the fact that the two knowledge sources exist.*	20	7.36	5	1.19	17.41**	2.63
PA 15	Initiator – Exemplified – Hinge – Indicator – Exemplification-RS e.g. *For discussions on difficulties in defining interruption, see for example, Tannen 1989; 1990: 188 ff.; Hopper 1992: 133).*	18	6.62	7	1.66	11.03*	2.00
PA 16	Researcher – Hinge – Exemplification-SC – Indicator – Exemplified e.g. *We then draw on the data in order to illustrate the argument.*	11	4.05	18	4.27	0.02	−0.08

Table 11 (*continued*)

No.	Pattern	TP1		TP2		LL	LR
		Raw	Normed	Raw	Normed		
PA 17	Researcher – Indicator – Exemplified – Exemplification-SC e.g. *We illustrated this metaphor by applying it to Allie's case.*	8	2.94	23	5.46	2.47	−0.89
PA 18	Initiator – Exemplification-SC – Indicator – Exemplified e.g. *As these examples illustrate, falsification is not a straightforward matter and is, in fact, extremely difficult, if not impossible to achieve.*	7	2.58	23	5.46	3.41	−1.08
PA 19	Exemplification-RS – Indicator – Exemplified e.g. Her studies illustrate how students participate …	5	1.84	15	3.56	1.80	−0.95
PA 20	Indicator – Exemplification-SC – Exemplified e.g. *As illustrated in Figure 1, the care plan consists of four parts.*	3	1.10	31	7.36	16.18**	−2.74
PA 21	Exemplified – Hinge – Exemplification-SC – Indicator e.g. *Results from this operation could readily be used to confirm gender stereotypes, as Table 1 illustrates.*	1	0.37	13	3.09	7.61	−3.07
Total		**5,936**	**2,184.34**	**9,558**	**2,268.42**	**5.24**	**−0.05**

Based on the analyses, the following observations can be tentatively drawn, although it should be cautioned that these observations may only apply to Linguistics research articles. The analyses suggest that the overall use of exemplification has stayed relatively steady across the two time periods. This indicates that, on the one hand, exemplification is indeed a prominent feature of academic discourse (Hyland, 2007) and, on the other, emphasis on the important role of exemplification in academic writing has not changed much over time.

Nevertheless, the thorough and systematic investigation uncovers some notable subtle changes. For example, patterns 1 and 2 are both used significantly more frequently in TP2. Since the two patterns are especially simple and explicit, their higher frequency in TP2 suggests that writers nowadays tend to exemplify in a simpler and more explicit way. This tendency is further supported by two additional pieces of evidence discussed below.

One is that some relatively complex patterns such as 5, 14, and 15 are used significantly less frequently in TP2. The other is that writers in TP2 appear to prefer active voice over the passive alternative, as evidenced in the significantly lower frequency of Pattern 3 and significantly higher frequency of Pattern 7 in TP2. Patterns 3 and 7 are both realized by the two performative verbs *exemplify* and *illustrate*, and the difference between them lies in the voice only. This trend aligns with the observations drawn in previous studies that passives are decreasingly used in scientific discourse in general (Seoane, 2006, 2013). Studies have also suggested that the decreasing use of passives reflects the emphasis on clarity in academic writing. For example, Hyland and Jiang (2019a) argue that,

> … there is now great pressure towards making arguments as clear as possible. One way to make oneself heard in the increasing clamour of academic competition is to present information with clarity so that it can be read easily and rapidly. Passives, however, slow down the efficient processing of information.
>
> (Hyland & Jiang, 2019a, p. 50, p. 50)

In addition, the increasing emphasis on simplicity and clarity, as reflected in the use of exemplification, may be attributed to "the emergence of AL [Applied Linguistics] as a scientific interdisciplinary field of inquiry" (Hyland & Jiang, 2019a, p. 82). More specifically, the journal *Applied Linguistics* covers a wide range of topics and its readership includes researchers from various subfields of Linguistics and possibly other related humanities and social science disciplines such as Psychology and Education. Speculatively, authors of this journal are aware of the interdisciplinary nature of the field and the journal's readership and thus attempt to present their arguments or observations as explicit as possible.

As discussed in Section 4.2, academic writers normally perform the discourse act of exemplification by presenting either relevant studies or sub-categories in academic discourse, corresponding to the distinction made between Exemplification-RS and Exemplification-SC in our analyses. Another question worth addressing, then, is whether academic writers tend to exemplify more in one way or the other and whether their tendency has changed over time. To address this question, we examined the distribution of patterns involving Exemplification-SC and Exemplification-RS across the two time periods. The results are shown in Table 13.

Table 13
Distribution of patterns with Exemplification-SC and Exemplification-RS.

Types of pattern	TP1		TP2		LL	LR
	Raw	Normed	Raw	Normed		
Patterns with Exemplification-SC	4,893	1,800.54	7,494	1,778.57	0.45	0.02
Patterns with Exemplification-RS	1,043	383.81	2,064	489.85	42.27**	−0.35

The results show that in both time periods academic writers of Linguistics tend to exemplify by presenting sub-categories more frequently than by presenting relevant studies. This suggests that the major function of exemplification in Linguistics research articles is to provide additional information to specify a superordinate category or an abstract concept by presenting subordinate ones and that this major function has not changed over time.

However, Table 13 also shows that academic writers tend to exemplify much more frequently by presenting relevant studies in recent years, as indicated by the significantly higher frequency of patterns of the type Exemplification-RS in TP2 than in TP1. One reason for this increase may be that the average length of the literature review section in research articles in TP2 is longer than that in TP1, as a result of the increase of the average length of each article (9,435.84 vs. 12,768.21; see Table 5) and the larger number of relevant studies to be reported (Hyland & Jiang, 2019a, 2019b). More notably, since the presence of relevant studies generally coincides with citations (e.g., … *much work is devoted to investigating the relationship between universals and transfer in L2 acquisition (e.g. Gass 1979; 1984; Zobl 1980; 1984; Anderson 1983)*), the increased use of Exemplification-RS patterns in TP2 reinforces Hyland and Jiang (2019a, 2019b) observation that citations have increased substantially over the past 50 years (especially in the field of Linguistics).

Among the patterns of the type Exemplification-RS, the increase in frequency was particularly notable for Pattern 2. In terms of citations, this pattern roughly corresponds to what has been referred to as 'parenthetical' (Swales, 2014) or 'non-integral' citations (Hyland, 1999; Hyland & Jiang, 2019a). Briefly, such citations usually make references to existing studies in parentheses, as illustrated in Example (13) and (14). They are normally used, for example, to direct readers to other sources and to save space (see Harwood (2009) for more discussion).

Ex. 13. *This pattern of development parallels that reported in other studies (e.g. Cazden et al., 1975; Wode, 1976).*

Ex. 14. *A number of quite complex models have been proposed to account for the range of situational factors that are involved (e.g. Crystal & Davy, 1969; Brown & Fraser, 1979).*

The observation that Pattern 2 has been increasingly used in TP2, then, may offer an additional explanation for Hyland and Jiang (2019a, 2019b) finding that non-integral citations have increased overwhelmingly. That is, non-integral citations have increased partly because academic writers nowadays tend to exemplify more frequently by presenting relevant studies (with the aim to keep the argument on track).

To recapitulate, the current investigation shows that, while the overall frequency of exemplification in Linguistics research articles has remained relatively stable over time, some significant changes have occurred in the specific ways in which academic writers exemplify. In particular, academic writers of Linguistics nowadays tend to exemplify in simpler and more explicit ways; and while they continue to exemplify primarily by presenting subcategories, they now also use relevant studies to illustrate arguments/viewpoints more frequently than before.

These changes in academic discourse are, arguably, responses to changes in social-cultural contexts. Specifically, the increasing emphasis on simplicity and clarity may be a result of the interdisciplinary turn in academic research as well as the trend that "human beings are adapting and using their languages in a simplified manner" (Lei & Wen, 2020, p. 12). In the case of the increase of exemplifying by presenting relevant studies in TP2, one particular factor that could account for this is the increasing pressure for academics to publish more and the ever growing volume and accessibility of research publications. In consequence, the increase of patterns with Exemplification-RS echoes the fact "that there is much more research to report and that referring to it is much easier now as a result of electronic access to source texts" (Hyland & Jiang, 2019a, p. 88).

5. Conclusion

In this primarily methodological study, we have proposed a local grammar approach to diachronic studies of discourse acts in academic texts and demonstrated the approach with a case study investigating diachronically exemplification in Linguistics research articles. Our investigation shows that, although the frequency of exemplificatory markers has remained relatively stable over time, that of the local grammar patterns associated with exemplification has changed in significant ways. The methodological implications are that local grammars offer a way to reliably quantify the realizations of discourse acts and, more notably, that local grammars are useful for unveiling current trends and exploring changes in language use, even when the lexicogrammatical forms have not changed significantly. The proposed local grammar approach can therefore be regarded a viable method to facilitate diachronic studies of discourse acts in academic texts and, by extension, in other kinds of specialized texts, complementing studies investigating diachronically other lexicogrammatical features in academic texts.

Provided the potential of local grammars in quantifying the ways in which a given discourse act can be realized, as demonstrated in the present study, a further methodological implication is that the proposed local grammar approach can also be applied to explore disciplinary variation by examining the realizations of discourse acts across disciplines. More research employing the local grammar approach to investigate diachronic changes and disciplinary variation in the realizations of diverse discourse acts in academic texts would thus be welcome and worthwhile.

Findings of this type of local-grammar-based investigations into academic discourse could have useful implications for EAP and ESP pedagogy (see also Zhang & Su, 2021). For example, our finding that exemplification remains a prominent feature

of research articles over time confirms the fundamental importance of exemplification in academic writing and the need to teach exemplification to novice or L2 English academic writers. In addition, the observation that writers nowadays tend to perform exemplification more explicitly suggests that it would be necessary and useful to teach novice or L2 English academic writers the explicit ways in which exemplification can be performed. To this end, the inventory of the lexicogrammatical variations and their associated local grammar patterns generated in the present study would constitute a useful pedagogical resource that can be applied to develop materials for the teaching of discourse acts in academic writing (Su & Zhang, 2020).

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Appendix. Patterns with relatively low frequency in both corpora

No.	Pattern	1980–1994		2000–2014	
		Raw	Normed	Raw	Normed
PA 22	Researcher – Indicator – Exemplified e.g. *We will now illustrate the primes of the model and then discuss its implications.*	9	3.31	7	1.66
PA 23	Exemplified – Indicator – Hinge – Exemplification-RS e.g. *The minimum cut-off point from which they are omitted must be reported (for example, see Goldman-Eisler 1961b:220).*	8	2.94	5	1.19
PA 24	Initiator – Indicator – Exemplification-RS e.g. *It has been hypothesized (e.g. by Tarone 1983) that interlanguage, like any other natural language, is systematically variable.*	7	2.58	4	0.95
PA 25	Exemplified – Exemplification-RS – Indicator e.g. *His more recent work reflects a different stance (Adjemian 1981; for example).*	4	1.47	2	0.47
PA 26	Researcher – Hinge – Indicator – Exemplified – Exemplification-SC e.g. *We have already seen an example of the latter in (6b), in which Peter accompanies …*	4	1.47	8	1.90
PA 27	Initiator – Hinge – Indicator – Exemplification-RS e.g. *It has been argued (see, for example, Rudolph 1989) that parallels in underlying meaning are not necessarily marked in the surface text.*	4	1.47	0	0.00
PA 28	Researcher – Hinge – Indicator – Exemplified e.g. *I shall finally provide an example of how the means analysis can be used to investigate …*	3	1.10	4	0.95
PA 29	Exemplification-SC – Researcher – Hinge – Indicator – Exemplified e.g. *In (7b) we have an example of an expansion, and in (7c) there is an example of a rephrasing as a response.*	3	1.10	2	0.47
PA 30	Researcher – Indicator – Exemplification-SC – Exemplified e.g. *We illustrate in the second part of the paper how we have done this.*	3	1.10	0	0.00
PA 31	Exemplification-RS – Hinge – Indicator – Exemplified e.g. *Blackledge and Creese (2008) is an example of an ethnographic study that provides interactional context through its extracts.*	2	0.74	3	0.71
PA 32	Exemplified – Indicator – Hinge – Exemplification-SC e.g *….cohesion can also indicate the scope of the clause relation where the two matched elements are non–adjacent (e.g. see M.3 discussed in 3.5 above).*	2	0.74	2	0.47
PA 33	Exemplified – Hinge – Researcher – Indicator – Exemplification-SC e.g. *Writers took this component of interpersonal power into account in framing compliments, as I will illustrate at the conclusion of the syntax section below.*	2	0.74	0	0.00
PA 34	Initiator – Indicator – Exemplification-RS – Supporting Statement e.g. *It should be mentioned that, for example, Verhallen and Schoonen (1993) also carried out a statistical analysis of their data, …*	0	0.00	4	0.95
PA 35	Indicator – Exemplified – Exemplification-SC e.g. *An example of acrostics for 'JOY': Jump Out of Your sorrow.*	0	0.00	2	0.47
PA 36	Initiator – Indicator – Exemplified – Exemplification-RS e.g. *For an example of the curricular inclusion of heritage speakers as tutors for L2 learners, see Quintanar-Sarellana* et al. *(1997).*	0	0.00	2	0.47
PA 37	Initiator – Exemplification-RS – Indicator – Exemplified e.g. *See Mondada and Doehler (2004) for an example of classroom-based research and Brouwer …*	0	0.00	2	0.47

Statement

The study reported in this paper has not been published previously or submitted for consideration in any other journals.

Author contributions

Hang Su: Conceptualization; Data analysis; Funding acquisition; Methodology; Project administration; Writing – original draft; Writing – review & editing. **Yuqing Zhang:** Data collection; Data analysis. **Xiaofei Lu**: Statistics; Writing – review & editing.

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Declaration of competing interest

None.

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